

# Module 4: Computational Complexity, Cryptography & NP

## CS 305: Theory of Computation (Weeks 13–15)

Department of Computer Science

Ghana Institute of Management and Public Administration (GIMPA)

Level 300 Undergraduate

# Module 4 Roadmap: Weeks 13 to 15

## Theoretical Foundations

- **Wk 13:** Asymptotic Growth & Class P
- **Wk 14:** Class NP, Cook–Levin Theorem & Polynomial-Time Reductions ( $\leq_P$ )
- **Wk 15:** Course Synthesis, P vs. NP Millennium Prize & Intro to BQP

## Research Literature & System Spotlights

- **Wk 13:** Scalability:  $\mathcal{O}(n^2)$  vs.  $\mathcal{O}(2^n)$  in Production
- **Wk 14: TACAS 2008:** Industrial SAT/SMT Solvers (Z3) & **USENIX 2016:** Post-Quantum Crypto
- **Wk 15: SIAM 1997:** Shor's Quantum Factoring Algorithm

## Week 13: Measuring Time Complexity (Sipser Ch. 7.1)

### Definition: Time Complexity

Let  $M$  be a deterministic Turing machine that halts on all inputs. The **running time** or **time complexity** of  $M$  is the function  $f : \mathbb{N} \rightarrow \mathbb{N}$ , where:

$f(n)$  = maximum number of steps that  $M$  uses on any input of length  $n$ .

### The Class P (Polynomial Time)

$$P = \bigcup_{k \geq 0} \text{TIME}(n^k)$$

**The Cobham-Edmonds Thesis:** P is the mathematical class of problems that are **realistically solvable on computers** (tractable).

## Week 13 Spotlight: Polynomial vs. Exponential in Production

| Time Complexity                       | $n = 10$                       | $n = 50$                        | $n = 100$                        | $n = 1000$                     |
|---------------------------------------|--------------------------------|---------------------------------|----------------------------------|--------------------------------|
| $\mathcal{O}(n \log n)$ (MergeSort)   | $3.3 \times 10^{-8} \text{ s}$ | $2.8 \times 10^{-7} \text{ s}$  | $6.6 \times 10^{-7} \text{ s}$   | $1.0 \times 10^{-5} \text{ s}$ |
| $\mathcal{O}(n^2)$ (QuickSort worst)  | $1.0 \times 10^{-7} \text{ s}$ | $2.5 \times 10^{-6} \text{ s}$  | $1.0 \times 10^{-5} \text{ s}$   | $1.0 \times 10^{-3} \text{ s}$ |
| $\mathcal{O}(n^3)$ (Matrix Multiply)  | $1.0 \times 10^{-6} \text{ s}$ | $1.25 \times 10^{-4} \text{ s}$ | $1.0 \times 10^{-3} \text{ s}$   | $1.0 \text{ s}$                |
| $\mathcal{O}(2^n)$ (Brute Force 3SAT) | $1.0 \times 10^{-6} \text{ s}$ | <b>35.7 years</b>               | $4 \times 10^{13} \text{ years}$ | <b>Heat death</b>              |

### Production Insight

The boundary between  $\mathcal{O}(n^k)$  and  $\mathcal{O}(2^n)$  is not merely an academic definition; it is the physical dividing line between what software can execute in real time vs. what will never terminate.

# Week 14: The Class NP (Nondeterministic Polynomial Time)

## Two Equivalent Definitions of NP (Sipser Def 7.19)

- 1 **Nondeterministic TM Definition:**  $NP = \bigcup_{k \geq 0} NTIME(n^k)$ . (Languages decidable in polynomial time by an NTM).
- 2 **Polynomial Verifier Definition:**  $L \in NP \iff$  there exists a deterministic polynomial-time verifier  $V(w, c)$  such that:

$w \in L \iff \exists$  certificate/witness  $c$  with  $|c| \leq \text{poly}(|w|)$  such that  $V(w, c)$  accepts.

## Intuition: Solving vs. Checking

- **Sudoku / 3SAT / Password Cracking:** Extremely hard to *find* a solution from scratch ( $\mathcal{O}(2^n)$ ).
- But once someone gives you a candidate solution, it takes only  $\mathcal{O}(n)$  **time to verify!**

# Week 14: Polynomial-Time Reductions & The Cook–Levin Theorem

## Polynomial-Time Reduction ( $A \leq_P B$ )

Language  $A$  polynomial-time reduces to  $B$  if there exists a polynomial-time computable function  $f : \Sigma^* \rightarrow \Sigma^*$  such that:

$$w \in A \iff f(w) \in B$$

## NP-Completeness Definition

A language  $B$  is **NP-complete** if:

- 1  $B \in \text{NP}$ , and
- 2 Every  $A \in \text{NP}$  satisfies  $A \leq_P B$ .

## The Cook–Levin Theorem (1971)

$\text{SAT} = \{ \langle \phi \rangle \mid \phi \text{ is a satisfiable Boolean formula} \}$  is NP-complete.

**Karp's Reductions:**  $3\text{SAT} \rightarrow \text{CLIQUE} \rightarrow \text{VERTEX-COVER} \rightarrow \text{SUBSET-SUM}$ .

## Industrial SAT/SMT: TACAS 2008

**De Moura & Bjørner (Microsoft Research),**  
“Z3: An Efficient SMT Solver,” **TACAS 2008.**

- CDCL solvers solve NP-complete formulas with **1,000,000+ variables!**
- Used by Amazon AWS and Intel for formal verification of cloud infrastructure and silicon chips.

## Post-Quantum Crypto: USENIX 2016

**Alkim et al., “Post-Quantum Key Exchange: A New Hope,”** **USENIX Security 2016.**

- If  $P = NP$ , all public-key crypto breaks!
- NIST standardizes Lattice Cryptography (Kyber/Dilithium) based on the worst-case hardness of high-dimensional lattice problems.

# Week 15 Research Spotlight: Quantum Complexity Class BQP & Shor's Algorithm

Seminal Reading: SIAM J. Comput. 1997 / IEEE FOCS 1994

**Peter Shor (MIT / Bell Labs)**, "*Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*," **SIAM Journal on Computing** 1997.

## The Quantum Complexity Class BQP

- Bounded-error Quantum Polynomial time (BQP).
- $P \subseteq BQP$ .
- Prime Factorization is in BQP (runs in  $\mathcal{O}(n^3)$  quantum time via QFT), but not known to be in P.

## Does Quantum Solve NP-Complete?

- **General Belief:**  $NP \not\subseteq BQP$ .
- Quantum computers speed up search (Grover's  $\mathcal{O}(\sqrt{N})$ ), but are **not** believed to solve NP-complete problems in polynomial time!

# Course Grand Synthesis: What Have We Learned?

| Model / Grammar    | Memory Mechanism                    | Language Class           | Real-World System                 |
|--------------------|-------------------------------------|--------------------------|-----------------------------------|
| DFA / NFA          | Finite States (No stack)            | Regular                  | Lexers, ReDoS, Firewalls          |
| Pushdown Automaton | LIFO Stack                          | Context-Free             | Compilers, LLM JSON Masks         |
| Turing Machine     | Infinite Read/Write Tape            | Decidable / Recognizable | General Computers, CoT LLMs       |
| Complexity Classes | Bounded Time ( $\mathcal{O}(n^k)$ ) | P vs. NP                 | Crypto, SAT Solvers, Optimization |

## Final Word to GIMPA Level 300 Graduates

You now possess the mathematical language that describes every computing device that has ever existed, and every computing machine that will ever be built.